Mathematical Foundations: A Teaching Companion

Six plain-language lessons with worked toy examples
Emission-Trajectory ML Project

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Abstract

This document re-teaches the mathematics of the project's foundations document, `main.pdf` in the `math` folder (31 numbered equations), as six plain-language lessons, each built around a small worked example that can be computed by hand and re-used in front of an audience. It is a companion for teaching and study, not a replacement for the formal document: every lesson cites the equation numbers of `main.pdf`, and the formal statements there remain authoritative. The lessons follow the order of the foundations document: evaluation metrics, baselines, ordinary least squares and Ridge, trees and boosting through XGBoost, uncertainty of a skill score, and error decomposition. Each lesson ends with teach-back questions; brief model answers are collected in the appendix. Prepared with Claude (Anthropic) from live tutoring sessions in August 2026.

Contents

1 How to use this document

Each lesson has the same shape: the concept in plain words, the equations with every symbol explained, one small worked example with numbers you can verify by hand, a figure showing the mechanism, the trap that catches students, and teach-back questions. The toy numbers are invented for clarity and are labeled as such; every claim about the real project is tied to the frozen results in `results/model_comparison.csv` and the Session 6 tables. Read with `main.pdf` open beside this document; equation numbers cited here refer to it.

2 Lesson 1: The referee. Five metrics are five different judges

2.1 Concept

A single miss is easy to describe: the forecast said 970, reality was 1000, the miss is 30. The mathematics becomes interesting only when hundreds of misses must be summarized in one number, because then someone must decide whose misses count more. Every metric in the foundations document (Equations 1 to 5) is a different answer to that one question, and none of the answers is neutral.

2.2 The pocket example

Five countries, one year, one prediction each from the model and from persistence (the rule "next year equals this year"). Persistence's prediction is last year's value, so its miss is the actual year-over-year change. The numbers are invented for teaching.

Country	Last year	Actual	Model predicts	Model miss	Persistence miss
A (very large)	965	1000	970	30	35
B (large)	105	100	104	4	5
C (mid)	53	50	48	2	3
D (small)	12.5	10	11	1	2.5
E (near zero)	2.5	0.5	1.5	1	2

2.3 The five judges

MAE (Equation 1), the average miss. $\mathrm{MAE} = (30 + 4 + 2 + 1 + 1)/5 = 7.6$. Of the 38 total error units, 30 come from country A alone, which is 79 percent of the verdict decided by one country. MAE is a democracy in which votes are weighted by tonnage. This is a choice, not a flaw, and it is why the project's real MAE numbers are dominated by the largest emitters.

RMSE (Equation 2), the disaster detector. $\mathrm{RMSE} = \sqrt{(900 + 16 + 4 + 1 + 1)/5} = \sqrt{184.4} \approx 13.6$. Squaring before averaging makes a miss of 30 contribute 900, so large misses count quadratically extra. The useful diagnostic is the ratio to MAE: if all five misses were identical, RMSE would equal MAE; here the ratio is about 1.8, which signals that the error budget sits in a fat tail of a few large misses. In the real project, `hgb_level` showed validation MAE 18.9 against RMSE 142.7, and that ratio alone indicated catastrophic misses on giants before any per-country table was opened.

MedianAE (Equation 3), the typical country's experience. Sort the misses (1, 1, 2, 4, 30) and take the middle one: 2. Country A's miss could grow from 30 to 3000 and this number would not move. The same model is "off by 7.6 on average" and "off by 2 for the typical country," and both statements are true; the gap between them measures how concentrated the errors are [10].

MdAPE (Equation 4), the miss relative to the country's own size. Divide each miss by the country's own level: A scores $30/1000 = 3$ percent, B 4 percent, C 4 percent, D 10 percent, and E explodes to $1/0.5 = 200$ percent because the denominator is nearly zero. Two teachable events occur at once. First, the ranking reverses: A has the largest absolute miss and the smallest percentage miss. Second, E shows why the threshold $\tau = 1$ Mt is part of the metric's definition rather than optional cleaning: with E excluded, MdAPE is the median of (3, 4, 4, 10), which is 4 percent. In the real test set, the same reversal appears with Russia (a top-10 absolute miss at the sample's lowest percentage error) against Pakistan and Vietnam.

Skill (Equation 5), error relative to doing nothing. Persistence's misses sum to 47.5, so its MAE is 9.5, and

$$\mathrm{Skill} = 1 - \frac{7.6}{9.5} = +0.2.$$

The reading: the model removed 20 percent of the error that "assume no change" would have made. Zero means the model added nothing; one is the unreachable perfect forecast; negative skill is a loss and must be reported as one.

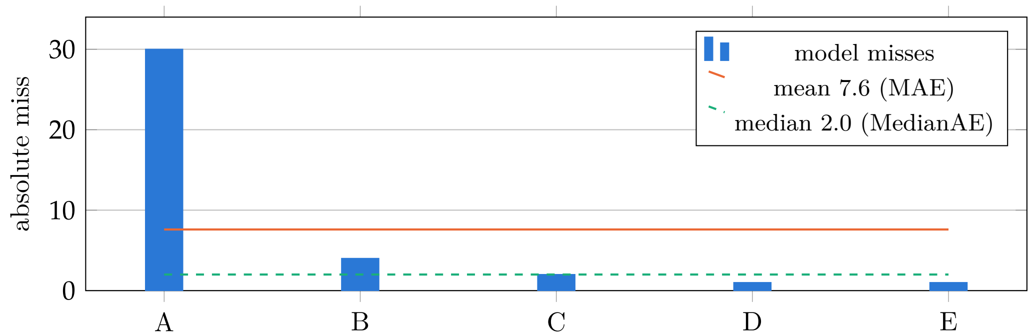

Figure 1: The five toy misses. One tall bar drags the mean to 7.6 while the median stays at 2.0 with the typical country.

2.4 Two fairness rules

The level and delta parameterizations (Equation 6): a model may be asked either "what will the level be" or "how much will it change," but if it answers the second (say a predicted change of +3 from a level of 120), the level $120 + 3 = 123$ is reconstructed and scored. The parameterization changes the question asked, never the quantity judged. The same-rows rule: every model is scored on the identical set of rows, because a model scored only on easy rows wins by taking a different exam. Both rules are enforced in `evaluate.comparison_table`.

2.5 The trap

The instinct is to ask which metric is best. The project's own test set answers that this is the wrong question: `xgb_delta` wins MAE (9.345 against 11.004) while persistence wins MedianAE (1.367) on the same predictions and the same rows. Neither verdict is an error. The right question is which judge answers the question at hand: total misallocated megatons is MAE's question; "will this work for a typical country" is MedianAE's.

2.6 Teach-back questions

1. In the toy table, MAE is 7.6 and MedianAE is 2.0. Explain what each number describes and why they differ by a factor of almost four.
2. A colleague reports "our model achieved MAE 9.3, an excellent result." What single number do you demand before accepting the claim?
3. Why is the $\tau = 1$ Mt exclusion part of MdAPE's definition rather than optional cleaning? Use country E to show what would happen without it.
4. Using only MAE and RMSE from a comparison table, how would you detect that a model is failing catastrophically on a few giants?

3 Lesson 2: The opponent. Baselines and knowing in advance where each fails

3.1 Concept

A baseline is a forecast with zero fitted parameters: nothing in it is estimated from data, so nothing in it can overfit or be tuned to flatter itself. Its job is to define the bar. If a model with 17 features cannot beat a rule a student can compute on paper, the machinery added nothing.

3.2 Persistence (Equation 7)

Persistence predicts that next year equals this year. Its miss is therefore the data's own year-over-year change, which has two consequences. First, persistence's MAE equals the average absolute annual change of the series, so the bar every model must clear can be computed before any model exists. Second, beating persistence means predicting the direction and size of the change better than the guess "zero change," which is a real demand, not a formality.

Persistence is the optimal forecast under a random walk, a series whose next value is the current value plus an unpredictable step of mean zero [9]. It is strong for CO_2 specifically because emissions come from physical stock (power plants, vehicle fleets, industrial capital) that cannot change quickly: in this dataset the median absolute year-over-year change since 2010 is about 4.4 percent of the level (Session 1 EDA).

3.3 Linear trend, or drift (Equation 8)

Drift measures the average yearly change over the last five years and assumes one more year of exactly that. Two invented countries make the trade-off visible.

A sustained riser: emissions 80, 84, 88, 92, 96, 100 over six years. The five-year slope is $(100 - 80)/5 = +4$ per year, so drift predicts 104 and persistence predicts 100. If the actual value is 104, drift's miss is 0 and persistence's is 4.

A country that peaked: emissions 90, 94, 98, 102, then a turn: 100, 98. The five-year slope is $(98 - 90)/5 = +1.6$, still positive two years after the peak, because the window still contains the boom years. Drift predicts 99.6, persistence predicts 98, and the actual value falls to 95. Persistence misses by 3; drift misses by 4.6 and points in the wrong direction. A five-year window remembers old growth long after reality has moved on.

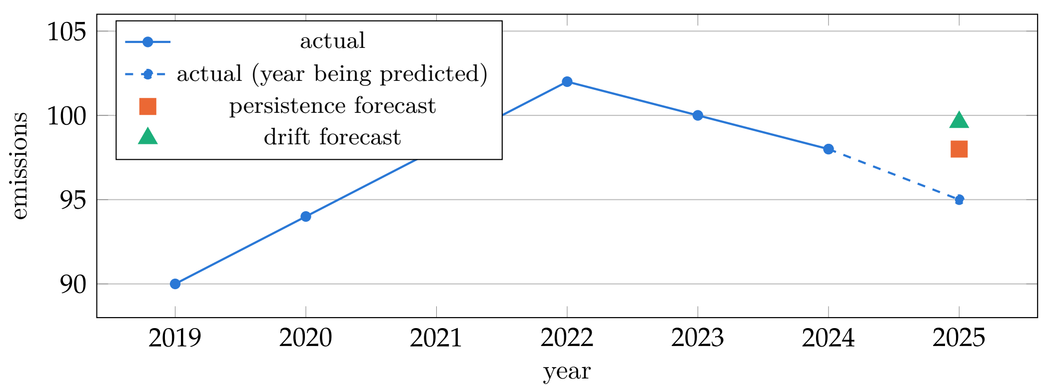

Figure 2: The peaked country. Drift's stale positive slope points upward two years after the turn; persistence misses less.

3.4 Bias direction, and the real echoes

For a trending country, persistence's misses lean in a predictable direction: it under-predicts a riser every year and over-predicts a decliner every year, with the sign of the bias tracking the country's own recent trend. This allows failure predictions before any computation: persistence systematically under-shot Qatar, which rose every year from 2020 to 2024 (101.3 to 125.8 Mt), and over-shot sustained decliners such as Germany.

On the real validation years, drift won in 39.2 percent of countries (the steady decliners) and persistence won the rest; overall MAE was 6.757 for persistence against 7.281 for drift, yet drift won MedianAE and MdAPE, with persistence's advantage concentrated in avoiding disasters among giants (RMSE 24.9 against 33.3). The COVID shock is the peaked-country example at scale, with a timing twist: persistence's worst test year was 2020 itself, while drift's was 2021, because the 2020 collapse had by then entered drift's slope window and poisoned it just as emissions rebounded.

3.5 The trap

Students meet baselines and conclude they are a formality. The project's actual result is the antidote: on validation no learned model beat persistence at all (best skill -0.017), and on test the best model's win was statistically indistinguishable from zero (Lesson 5). Any report of model MAE without a persistence bar has hidden the hardest question: what would doing nothing have scored?

3.6 Teach-back questions

1. Persistence's miss equals the country's actual year-over-year change. Explain why, and state what beating persistence therefore demands.
2. For a country whose emissions peaked in 2010 and have declined steadily since, which direction does persistence miss, and which baseline should win there? What event would flip the winner back to persistence?
3. Using the peaked-country example, explain mechanically why drift's worst real test year was 2021 rather than 2020.

4 Lesson 3: The first learner. Ordinary least squares and Ridge

4.1 The scoring machine, and what S(beta) means

We predict this year's emissions from one feature, last year's emissions, using three rows of invented data:

Row	Last year (feature)	This year (actual)
1	10	10.5
2	20	21
3	40	42

The model is: prediction $= \beta \times$ last year, where β is one weight to be chosen. Persistence is the special case $\beta = 1$ imposed by decree; a learned model lets the data choose β.

Define a scoring machine S: feed it one candidate weight, and it makes all three predictions, measures the misses, squares them, and adds them up. The notation $S(1)$ reads "S of 1," the machine's score for the input 1; the parentheses are function notation, not multiplication. Running the machine by hand:

$$S(1) = 0.5^2 + 1^2 + 2^2 = 5.25, \qquad S(0.95) = 1^2 + 2^2 + 4^2 = 21, \qquad S(1.05) = 0.$$

4.2 OLS (Equation 9): report the arg min

Feeding the machine many weights produces a bowl-shaped curve (Figure 3). Two different questions can be asked of the bowl. The min asks how low the curve goes (here 0, a height); the arg min asks at which weight it goes that low (here 1.05, a position). Equation 9 of the foundations document instructs: report the arg min. The output of least squares is a weight, not an error value.

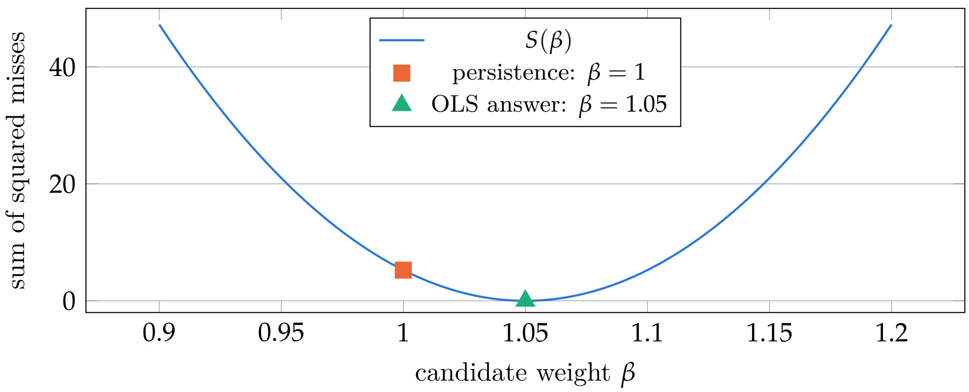

Figure 3: The bowl. The min is the height 0; the arg min is the position 1.05; least squares reports the position.

The closed form (Equation 11) is a division in disguise: the sum of feature times answer ($10 \times 10.5 + 20 \times 21 + 40 \times 42 = 2205$) divided by the sum of feature times feature ($100 + 400 + 1600 = 2100$), giving $\hat{\beta} = 2205/2100 = 1.05$. Read it as co-movement divided by self-movement. With 17 features the division becomes a matrix inverse, but the meaning is unchanged. In real data the bowl's bottom never touches zero; the toy rows were invented to fit perfectly.

4.3 Ridge (Equations 10 and 11): a price on weight size

The project's feature set contains near-twin columns (`co2_lag1`, `co2_roll5_mean`). When two columns carry almost the same information, OLS cannot decide how to split the weight between them: $+30$ on one twin and -29 on the other fits the training data about as well as 0.5 and 0.5, so the fitted weights swing wildly with tiny changes in the data. Ridge changes one thing: the scoring machine now charges for two items,

$$\text{ridge score} = (\text{sum of squared misses}) + \alpha \times (\text{sum of squared weights}),$$

where $\alpha \geq 0$ is the price level. Among weight combinations that fit equally well, Ridge picks the smallest ($0.5^2 + 0.5^2 = 0.5$ is far cheaper than $30^2 + 29^2 = 1741$), so near-twins share their weight instead of fighting over it. The addition of α inside the closed form also guarantees that the division always works [8]. In the one-feature example the best weight becomes $2205/(2100 + \alpha)$: raising α shrinks the weight toward zero, and at extreme α the model degenerates into predicting the same constant for everyone.

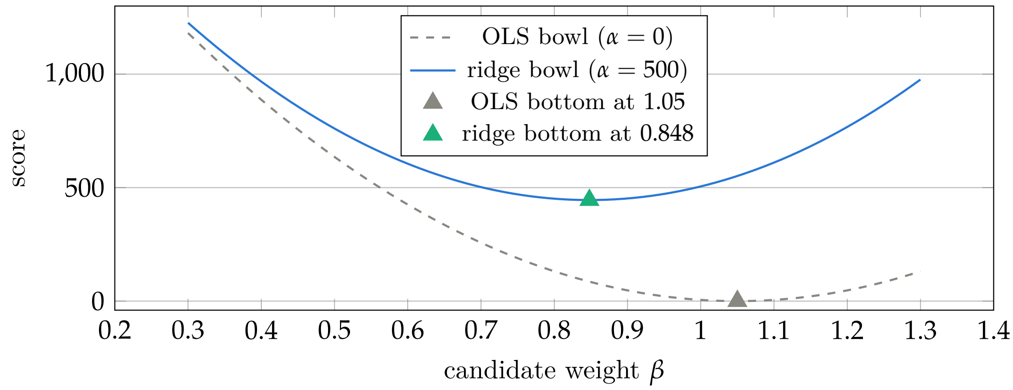

Figure 4: Adding the penalty bowl moves the combined bottom toward zero: the arg min slides from 1.05 to $2205/2600 = 0.848$ at $\alpha = 500$.

Two footnotes complete the section. The penalty charges every weight the same price per unit of size, so features must be standardized to comparable scales first, with the scaler fit on training rows only. And α is chosen by validation, not theory: Session 4 tried 0.1, 1, 10, 100 and validation MAE worsened as α grew (7.29, 7.53, 8.58, 9.95 for the level model), so the project froze $\alpha = 0.1$: a whisper of penalty, enough to steady the twins without distorting the signal.

4.4 Teach-back questions

1. Inside $S(\beta)$, what is being squared and summed, row by row?
2. In the bowl picture, which number is the min, which is the arg min, and which one is the fitted weight?
3. Persistence uses $\beta = 1$ and OLS found $\beta = 1.05$. What did OLS learn that persistence assumes away?
4. A colleague sets α to one million. What happens to the weights, and what forecast does the model degenerate into?

5 Lesson 4: The heavy machinery. Trees, bagging, boosting, XGBoost

5.1 Trees (Equation 12): staircases with a ceiling

A tree asks yes-or-no questions ("is last year's emissions 50 or more?") and predicts, within each resulting group, the group's average. Choosing the average is not arbitrary: it is the value minimizing the sum of squared misses within the group, the same bowl as always, applied per group. Two consequences follow. Trees need no feature scaling, because a threshold question cares only about the ordering of values, not their units. And a tree cannot predict outside what it has seen: every prediction is an average of training answers, so the prediction surface is a staircase, and the staircase has a top step (Figure 5).

5.2 Bagging (Equation 13) and boosting (Equations 14 and 15)

Bagging fits many trees, each on a bootstrap resample of the training rows, and averages their predictions: a parallel committee whose individual quirks cancel [1]. Random forests additionally

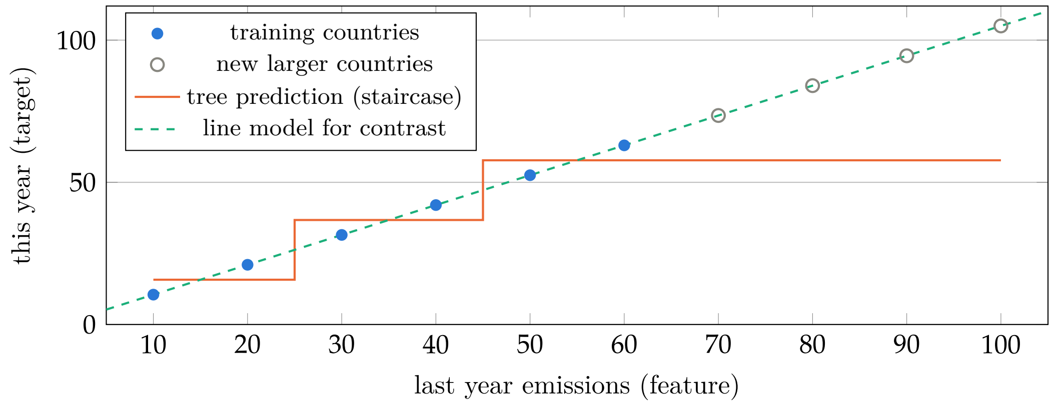

Figure 5: Inside the training range the staircase tracks the data; beyond it, the tree stays flat at its top step while the truth keeps rising.

restrict each split to a random subset of features so the committee members disagree more [2].

Boosting is instead a sequential relay: the model is a sum of many small trees built one at a time, and each new tree is trained on the leftovers (the residuals, actual minus current prediction) of the team so far [7]. With one country whose actual value is 100: tree 1 predicts 90 (leftover 10), tree 2 adds 7 (total 97), tree 3 adds 2 (total 99), tree 4 adds 0.7. Each tree is weak; the sum creeps toward the target (Figure 6). Scikit-learn's `HistGradientBoostingRegressor` is this machine with feature values binned into histograms.

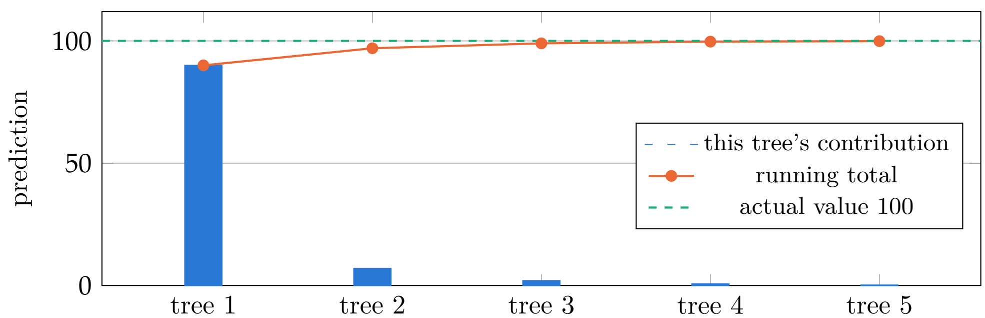

Figure 6: The boosting relay: shrinking corrections, a running total that creeps toward the target.

5.3 XGBoost (Equations 16 to 24): rooms, leftovers, and prices

XGBoost is the relay plus the idea from Ridge: complexity must be paid for [4]. The vocabulary, grounded in one concrete walkthrough. Freeze the relay mid-work, with four countries:

Country	Actual	Team predicts so far	Leftover
A	100	90	+10
B	60	46	+14
C	20	22	−2
D	12	16	−4

The new tree asks one question ("is last year's emissions 50 or more?"), which sorts A and B into one group and C and D into another. A room (the formal term is a leaf) is such a group, and each room hands one shared correction to its members. Plain boosting uses the room's average leftover: +12 for the first room, −3 for the second. XGBoost multiplies the average by $n/(n+\lambda)$, where n is the number of rows in the room: with $\lambda = 1$, the corrections become $12 \times 2/3 = +8$ and $-3 \times 2/3 = -2$ (Equation 22). This evidence shrink is Ridge's philosophy relocated into every room: a room holding two odd country-years is heavily damped, while a room holding 100 rows keeps almost its full correction (Figure 7). The dictionary:

Plain word	In the walkthrough	Formal term
Leftover	the column actual − predicted	residual
Room	the group $\{A, B\}$ or $\{C, D\}$	leaf
Correction	the room's one number, +8 or −2	leaf weight w
Shrink	the multiplier $2/(2+1)$	$n/(n+\lambda)$, Eq. 22
Rent	the entry fee a new split must pay	γ, Eqs. 16, 21
Pace	scale each correction by 0.10	learning rate η, Eq. 23

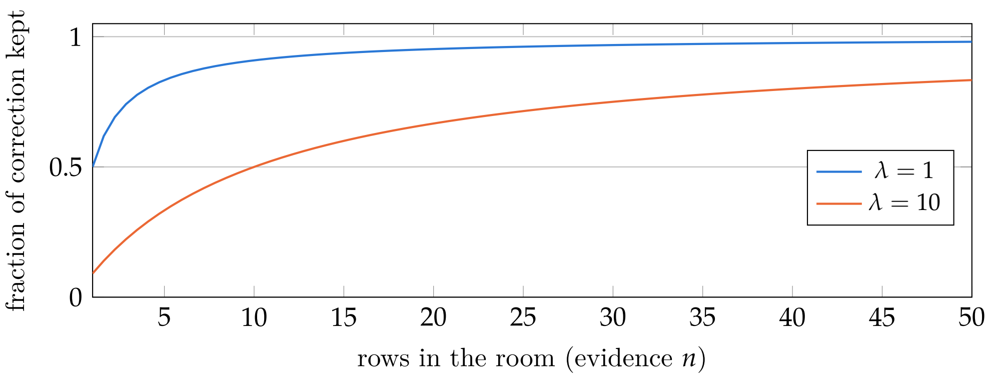

Figure 7: The evidence shrink $n/(n+\lambda)$. Thin-evidence rooms are damped hardest; well-evidenced rooms pass almost untouched.

Three more pieces complete the machine. The rent γ: a candidate split is scored by the improvement it brings minus γ (Equation 21), so a split improving the objective by 7 with rent 4 is made (net +3) while one improving by 2 is never born (net −2); this prunes memorization such as single-country rooms. The learning rate η: each finished tree enters the team scaled by η (Equation 23), so with the project's $\eta = 0.10$ the first room's +8 enters as +0.8 and the relay compensates by running many trees. Early stopping (Equation 24): trees are added while the validation MAE improves and the count is frozen at the best point M^* (Figure 8); the model is then refit on training plus validation rows with the count locked and early stopping disabled, so the test years never influence any choice. The project's frozen configuration: `xgb_delta`, depth 3, $\eta = 0.10$, $M^* = 173$ trees. A software footnote: the xgboost implementation reports split gains without the constant one half of Equation 21, which never changes which split wins; the foundations document verified this against xgboost 3.3.0 on a worked example.

5.4 Why trees require the delta parameterization

The staircase ceiling meets the test set: countries whose emissions rose above anything in training saturate a level-parameterized tree, which then systematically under-predicts every growing giant.

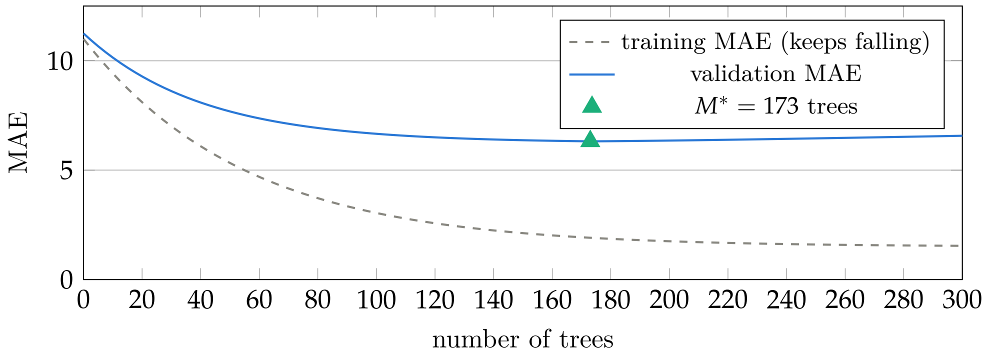

Figure 8: Early stopping, illustrative curve shapes. Training error falls forever; validation error bottoms out where memorization begins. The project's real frozen count is 173.

This is visible in the real results: `hgb_level` posted validation RMSE 142.7, and `xgb_level` was the one model that significantly lost to persistence on test (skill -1.834, $p = 0.005$). The delta parameterization sidesteps the ceiling: the tree predicts the bounded year-over-year change, well within training experience, and the level is reconstructed from the known current value. The unbounded part comes from the data; the tree handles only the bounded part.

5.5 Teach-back questions

1. A room contains three countries with leftovers +6, +9, and +9, and $\lambda = 1$. What correction does plain boosting hand out, and what correction does XGBoost hand out?
2. Ridge's α and XGBoost's λ are the same idea deployed in different places. What does each one shrink?
3. A student asks why not always use the line model, since it extrapolates. Give one honest sentence for each side.
4. Why does the project's protocol allow validation to choose the tree count but never the test years?

6 Lesson 5: Is the win real? Uncertainty of a skill score

6.1 Concept

The test result says `xgb_delta` has skill +0.151. But that number was computed on one particular sample: the 153 countries that exist, over five particular test years. Six heads in ten coin flips does not prove a biased coin; the question is how much the skill would wobble if the world had dealt a slightly different set of countries.

6.2 The loss differential (Equation 25), and why rows are not independent

For every row, subtract the model's miss from persistence's miss; a positive difference means the model won that row [5]. Four example rows expose the key problem:

Row	Persistence miss	Model miss	Difference d
China 2019	20	12	+8
China 2020	30	24	+6
Qatar 2019	4	5	−1
Qatar 2020	3	5	−2

China's rows are both positive; Qatar's are both negative. Rows travel in country packs: a model that understands China's dynamics wins all five of China's test years together. So China's five rows are one witness interviewed five times, not five independent witnesses. The test set has 762 rows but only 153 countries, and the honest amount of evidence is closer to 153. Treating rows as independent would overstate the certainty, which is why a row-level interval is wrong for this panel.

6.3 The country-clustered bootstrap (Equations 26 to 28)

The fix is a resampling game played at the country level [6, 3]. One round: put the 153 country names in a hat; draw 153 names with replacement (some countries repeat, some sit out); pool every test row of the drawn countries and recompute the skill (Equation 26). Repeat 2000 times. Because whole countries are drawn, each country's years stay together and the family-pack correlation is preserved. The spread of the 2000 replicate skills is the wobble. The 95 percent confidence interval is the range from the 2.5th to the 97.5th percentile of the replicates (Equation 27), and the p-value is twice the fraction of replicates on the wrong side of zero (Equation 28).

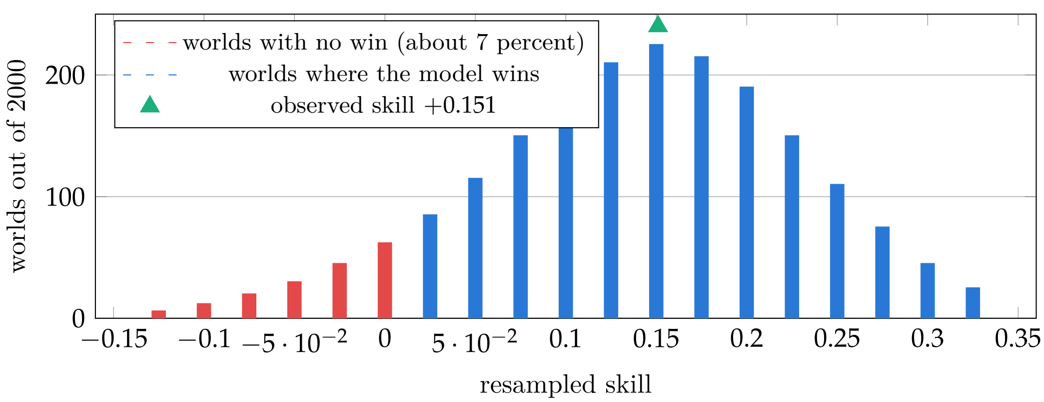

Figure 9: Illustrative sketch of the bootstrap distribution for the real headline result. The interval and p-value are the project's real numbers; the histogram shape is a sketch consistent with them.

6.4 The verdicts in honest English

Three from the real project. First, `xgb_delta` on test: skill +0.151, interval $[-0.018, +0.271]$, $p = 0.14$; the honest sentence is that the point estimate favors the model but the evidence cannot rule out a tie with persistence. Second, linear trend on validation: skill −0.078, interval $[-0.34, +0.20]$, $p \approx 0.7$; not "trend loses" but "trend and persistence are statistically indistinguishable on this sample." Third, `xgb_level` on test: skill −1.834, $p = 0.005$, the one settled verdict in the table, and it is a loss. The symmetry worth teaching: the skill score keeps a model honest against doing nothing, and the bootstrap keeps the skill score honest against luck.

6.5 Teach-back questions

1. A journalist writes "the model beat the baseline by 15 percent." Using the first verdict above, write the one-sentence correction.
2. Why are countries, not rows, resampled? What would a row-level resample destroy?
3. What does it mean that the interval $[-0.018, +0.271]$ contains zero?

7 Lesson 6: Where does the win live? Error decomposition

7.1 Group MAE and within-group skill (Equations 29 and 30)

No new machinery: slice the test rows into groups (years, countries, or emitter-size tiers), compute the ordinary MAE inside each group, and judge each group against persistence's error in that same group. Each tier gets its own honest contest with the bar set locally.

7.2 The trap this section exists to expose

Six invented countries in two tiers:

Tier	Rows	Persistence misses	Model misses
Giant	2	100, 100	60, 60
Small	4	2, 2, 2, 2	4, 4, 4, 4

Giant tier: model MAE 60 against 100, skill $+0.40$. Small tier: model MAE 4 against 2, skill -1.0; the model is twice as bad. Overall: persistence total error 208, model total 136, overall skill $1 - 136/208 = +0.35$. The model is worse for four of the six countries, the clear majority, yet the overall number looks excellent, because the error mass lives in the two giants. A single headline cannot distinguish "wins everywhere" from "wins where the tonnage is."

The real project mirrors the toy almost exactly (Figure 10): the headline test skill of $+0.151$ decomposes into giant tier (25 rows at 1000 Mt and above) skill $+0.325$, large and mid tiers approximately zero, and small tier -0.863, where every learned model loses to persistence. The deployment claim must be tier-scoped. By year, the same equation shows `xgb_delta` with positive skill in all five test years but earning most in calm 2019 and 2022, so the gain reads as steady-state structure rather than shock anticipation.

7.3 Permutation importance (Equation 31), and the twin caveat

To ask whether the model relied on feature j, shuffle column j across the evaluation rows, destroying its information while preserving its distribution, and measure how much the error rises [2]. The caveat follows from Lesson 3's twins: when two features carry overlapping information, shuffling one lets the model lean on the other, so both can score near zero even when their shared signal is essential. The real numbers show it: the five-year slope ranks first (2.889, unique directional information), while `co2_roll5_mean` scores just 0.098, not because the level family is useless but because it is redundant. Importances over correlated features describe group structure, never a per-feature verdict, and no feature is added or dropped in response, since the configuration is frozen post-test.

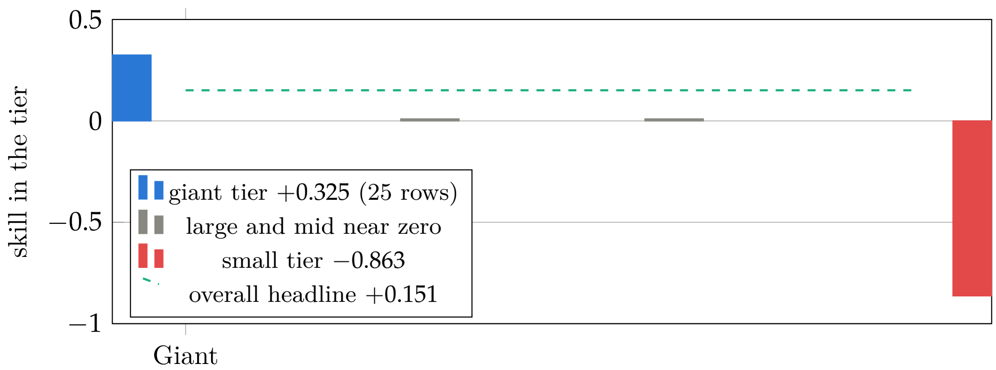

Figure 10: The real test-set decomposition. The overall headline sits between the tiers and describes none of them.

7.4 Teach-back questions

1. A ministry official from a small emitting country asks whether to adopt the model for national projections. Give the two-sentence honest answer using the tier figure.
2. Construct in words a situation where overall skill is positive while the model loses in most groups. What makes it possible?
3. Two features both score near zero permutation importance. Give two different explanations, and say what additional check separates them.

8 The whole document in six sentences

First, a miss can be counted five ways, and the choice of judge is itself a modeling decision; no metric is neutral. Second, persistence's error is the data's own year-over-year change, so the bar to clear is computable before any model exists, and for slow-moving CO_2 that bar is high. Third, OLS walks to the bottom of the squared-miss bowl, and Ridge adds a price on weight size, trading a small bias for stability when features are near-twins. Fourth, trees are staircases with a ceiling, boosting is a relay of corrections, and XGBoost is that relay with rent on rooms, an evidence shrink on every correction, and a validation-chosen stopping point; the ceiling is why the delta parameterization wins. Fifth, rows travel in country packs, so whole countries are resampled, and the honest headline is a point estimate whose interval includes zero. Sixth, the win lives almost entirely in 25 giant rows while small emitters are better served by no model at all, and importances over twins describe groups, not features.

A Model answers to the teach-back questions

Lesson 1. (1) MAE is the tonnage-weighted average miss and is dominated by country A; MedianAE is the middle miss and reports the typical country; they differ because the errors are concentrated. (2) The persistence MAE on the same rows: without the bar, 9.3 is uninterpretable. (3) Percentage error divides by the level, which explodes as the level approaches zero; without the threshold, country E alone would drag the metric toward 200 percent. (4) An RMSE far above MAE signals that the error budget is concentrated in a few large misses.

Lesson 2. (1) Persistence predicts last year's value, so its miss is exactly the change; beating it means forecasting the change better than guessing zero change. (2) Persistence over-predicts a decliner; drift should win; a sudden reversal (a rebound, a shock) flips the winner back to persistence. (3) In 2021 the 2020 collapse sat inside drift's five-year window, dragging its slope downward just as emissions rebounded; the window is a memory that a shock poisons for years.

Lesson 3. (1) The per-row misses, actual minus predicted. (2) The min is 0, the arg min is 1.05, and the fitted weight is the arg min. (3) That these countries grow about five percent per year. (4) All weights are pushed essentially to zero and the model degenerates into predicting the same constant for every country.

Lesson 4. (1) Plain boosting: the average, $+8$. XGBoost: $8 \times 3/4 = +6$. (2) α shrinks the regression's feature weights globally; λ shrinks each room's correction, in proportion to how little evidence the room contains. (3) The staircase captures interactions and thresholds no single line can, but it cannot predict outside the training range; the line extrapolates, but only along one straight direction. (4) Validation exists to make modeling choices; the test set exists to be touched once, so any choice influenced by it would turn the test into a second validation set.

Lesson 5. (1) "The model's error was 15 percent lower as a point estimate, but the 95 percent country-clustered interval includes zero, so a tie with the no-change baseline cannot be ruled out." (2) A country's years are correlated; row-level resampling would scatter them and understate the variance. (3) That among plausible worlds implied by the data, some show the model not winning; the gap is within sampling noise.

Lesson 6. (1) "For small emitters this model is worse than assuming no change, with tier skill -0.863; the gains are concentrated in the largest emitters, so I would not recommend it for your case." (2) A few groups with very large errors dominate the total while the model loses slightly in many small-error groups; concentration of error mass makes it possible. (3) Either the feature is useless, or it is a twin of another feature that covers for it when shuffled; shuffling the whole correlated group together separates the two cases.

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